Computer Science & IT



C Programming

String

Lecture No. 02



By- Abhishek Sir

Recap of Previous Lecture



- Topic — String
- Topic
- Topic
- Topic
- Topic

Slide

Topics to be Covered

#include<string.h>



- Topic
- Topic
- Topic
- Topic
- Topic

String function

Structure

strlen()
strcpy()
strcmp()
strcat()

string.h

Slide



String

String function

- strlen
- strcpy
- strcat
- strcmp



Strlen

The syntax of strlen() function in C is as follows:

strlen() it calculates length of string.

* strlen(const char *ptr)

* return value of strlen( ) . unsinged integer

* '\0' Not counted in length.

Slide



Strlen

Parameters

The strlen() function only takes a single parameter.

str: It represents the string variable whose length we have to find.

```
Count = 0
while(*ptr != '\0'){
    count++;
    ptr++;
```

```
char ch = 'A';
const char *ptr;
ptr = &ch;
```

```
Const char *ptr. }
```

using ptr ch value can't be update it will give error

Slide



Strlen

Return Value
This function returns the integral length of the string passed unsigned value.

\0 not included

"%lu"

↑

for Not having any warning

strlen("Hello") ⟹ this will return value 5

```
char ch[] = "parakram";
printf("%lu", strlen(ch));
```

8

Slide



String

```c
#include <stdio.h>
#include<string.h>

int main(){
    char a[4] = "string";
    printf("%lu", strlen(a));
    return 0;
}
```

length 4 — warning : exceeds character

4

GATE 2004

char *S = "string" : (ROM)

#Q Consider the following C program segment:

```
char p [20];
int i;
char *s = "string";

int length = strlen(s);

for (i=0 ; i<length; i++)

        p[i] = s[length - i];

printf("%s",p);
```

[d]

length = 6

The output of the program is

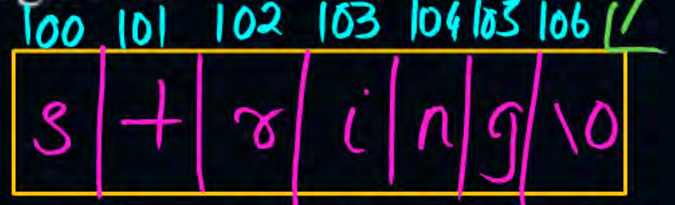
(a) gnirts

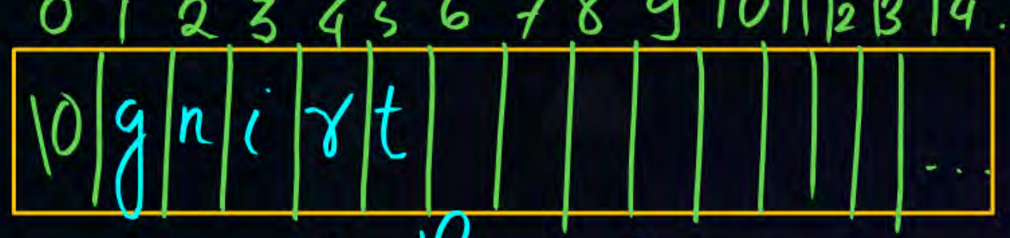
(b) string

(c) gnirt

(d) no output is printed

S 100





P[0] = S[6]

No output

("%s", p)

printf every thing till Null character found.

Slide



Strcpy

```
strcpy( char *destination, const char *ptr)
```

```
char a[20];
strcpy(a, "paakram");
```

Copy the string in first parameter string of second parameter.

Slide



Strcpy

```
strcpy( char *destination, const char *ptr)
```



Strcpy

Copy the character from source to destination

```
char a[100];

strcpy(a, “hello World”);

printf(“%s”, a);
```

String

```
#include <stdio.h>
#include<string.h>
int main(){
    char ch[20] ="parakram";
    strcpy(ch, "vijay");
    printf("%s", ch);
    return 0;
}
```

strcpy(char * Destination, const char * pto);

v	i	j	a	y	\0	a	m	\0

printf will print

vijay

p	a	r	a	k	r	a	m	\0

NULL character will be also copied



strcmp

prototype int strcmp(const char *ptr, const char *ptr,)

two strings are being compared

* if string are equal return value 0
* if two string are $S_1 < S_2$ the less than 0
* if two strings S_1 & S_2 are passed $S_1 > S_2$ the greater than 0 value return

Slide



strcmp

ASCII

A - 65

a - 97

'0' - 48

printf("%d", strcmp("abc", "abc"));

output = 0 $S_1 = S_2$

printf("%d", strcmp("Abc", "abc"))

65 97

S_1 S_2

-1 $S_1 < S_2$

printf("%d", strcmp("Abcdef", "abc"));

-1

printf("%d", S_1 S_2

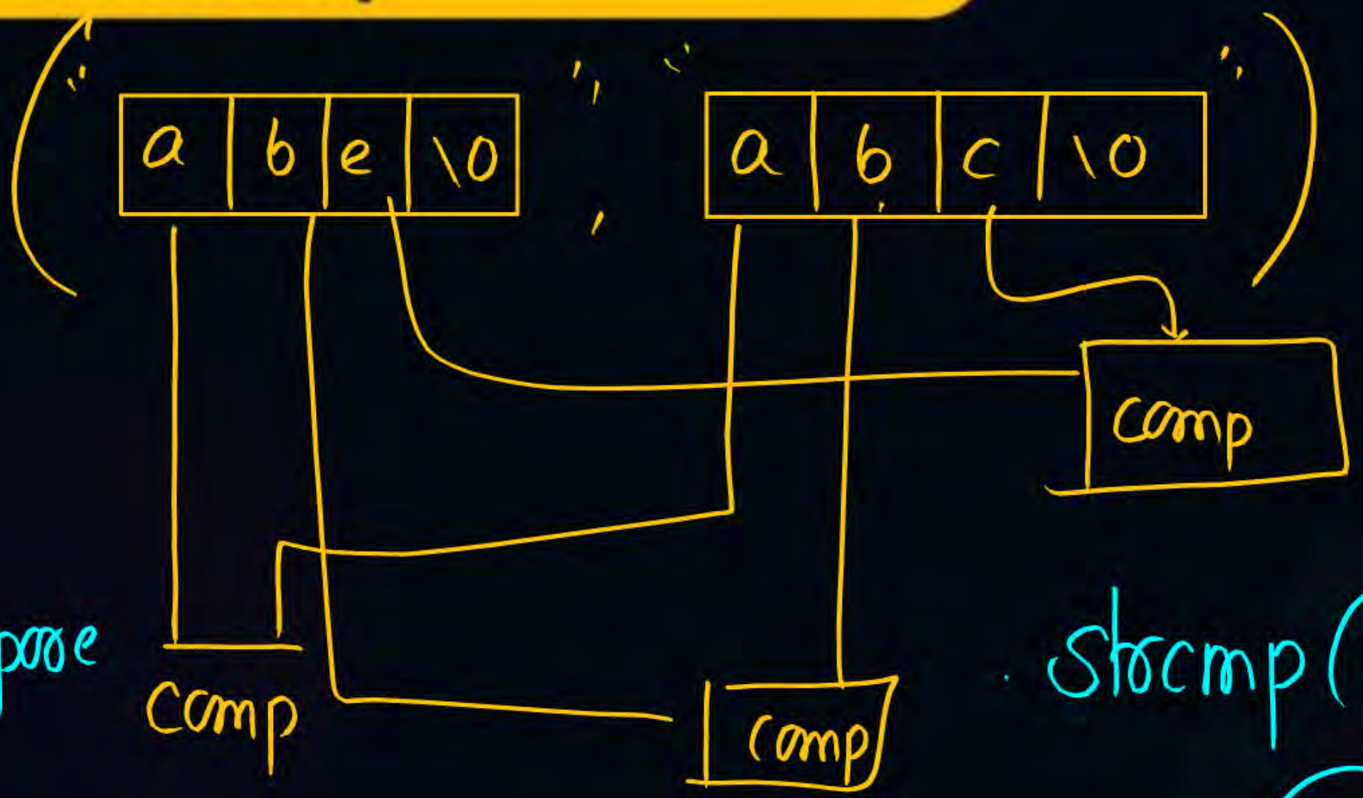
strcmp("abcd", "abc")

① $S_1 > S_2$

Slide

strcmp

Stocmp



Incremental way we compare the string

$S_1 < S_2$

strcmp("abc", "abcd")

-1

Slide



strcat

Concatenation

```
char *strcat(char *dest, const char *src);
```

Concatenation

destination string

const char *src

Source string

Strcat Concatenate Same string at the end of destination string



strcat

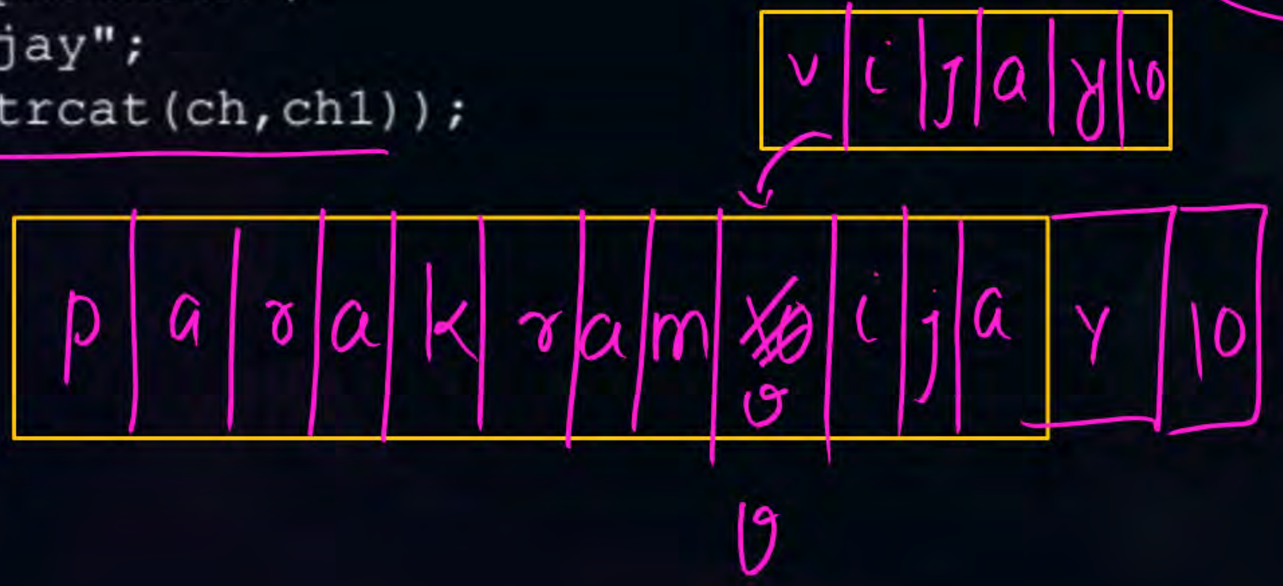
```
#include <stdio.h>
#include<string.h>
int main(){

    char ch[20] ="parakram";
    char *ch1 ="vijay";
    printf("%s", strcat(ch,ch1));
}
```

parakramvijay

output

return value char *



Slide



strcat

```
#include <stdio.h>
#include<string.h>
int main(){

    char ch[20] ="parakram";
    char *ch1 ="vijay";
    printf("%s", strcat(ch,ch1));
}
```



Structure

Array is collection Similar data type.

Structure is collection of disimilar data type.

user defined data :- Employ Data; ID :

Name :

phone No :

Address :

→ According to requirement we can create group of dissimilar Data type

Slide



Structure

```
struct A {
    int a;
    char ch;
};
```

New keyword **struct**

Member of structure

As if want to declare integer variable

```
int a;
```



Structure

for declaring structure variable

```
Struct A var;
```

Name of Structure

structure variable

Slide



Structure

How structure variable access member of structure.

"." Dot operator

```
Struct A var;

var.a = 10;

var.ch = 'A';
```

Name of structure variable.

- Member of structure.

Slide



Structure

Initialize the structure variable;

Struct A var = {10, 'A'}; As we initialize the array

GATE 2011

Q What does the following fragment of C-program print?

```
#include <stdio.h>
#include<string.h>
int main(){
    char c[]="GATEWALLAH";
    char *p =c;
    printf ("%s", p+ p [8] - p [1]);
}
```

0	1	2	3	4	5	6	7	8	9	10
G	A	T	E	W	A	L	L	A	H	\0

100

(A) Same string

(B) Different string

(C) TE WALLAH

(D) WALLAH

Same string

'0' 48

ASCII

value

zero character

P = 100

100

P[8] = *(100+8)

100+65 - 65

P[1] = *(100+1)

if('0') true

if(0) false

Slide

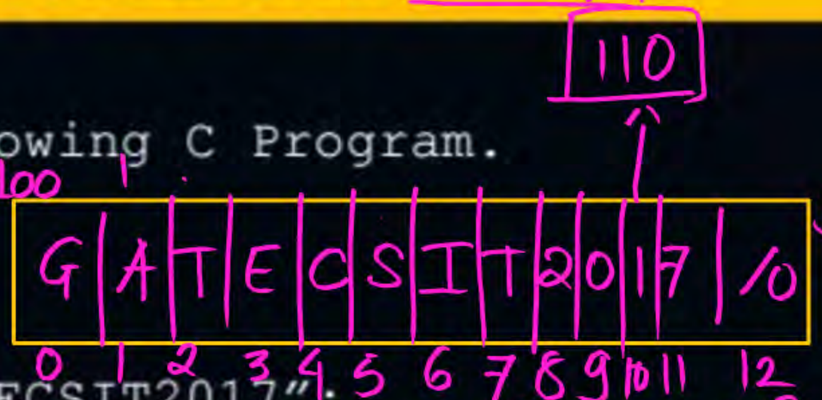


GATE ~~2004~~ 2017

```
#Q Consider the following C Program.
#include <stdio.h>
#include< string.h>
int main () {
      char* c = "GATECSIT2017";
      char* p = c;
      printf("%d", (int) strlen (c+2[p]-6[p]-1));
      return 0;
}
```

"lu",

The output of the program is (2) _____.



110

Stolen return unsinged

p[2] - p[6] -1

we are typecasting to integer

(2) > typecasting

a[i] = i[a] = *(a+i) = *(i+a) 2[p] 6[p]

T

(100 + 84 − 73 − 1)

T

Slide



2 mins Summary

- Topic
- Topic
- Topic
- Topic
- Topic

THANK - YOU

